A Survey on Representation Learning for Conversational Artificial Intelligence: From Similarity Preservation to Controllable Steering

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Abstract

Learning numerical representations that preserve semantic similarity has evolved from count-based distributional models to contextualized large language model (LLM) embeddings. Early methods such as Word2Vec (Mikolov et al., 2013a,b) and GloVe (Pennington et al., 2014) established that geometric proximity, measured by $\cos(e_{w_i}, e_{w_j})$, can encode semantic relatedness, but static vectors $e_w = f(w)$ suffer from polysemy and entanglement. Modern approaches address this by modeling $e_w = f_\theta(w|c)$ via masked language modeling (Devlin et al., 2019) and by explicitly optimizing sentence-level similarity through contrastive objectives (Reimers and Gurevych, 2019; Gao et al., 2021), leading to LLM-derived general-purpose embeddings (Jiang et al., 2024; BehnamGhader et al., 2024). This survey organizes these advances around a central challenge for conversational AI: detecting and understanding concepts in representation space to enable reliable steering. We show that while similarity preservation is now well achieved on benchmarks such as STS and MTEB (Muennighoff et al., 2023), isolating actionable concept directions v_C for interventions of the form $h' = h + \alpha v_C$ (Turner et al., 2023; Zou et al., 2023) remains difficult due to superposition and non-orthogonality (Elhage et al., 2022). We conclude with limitations and future directions toward disentangled, causally controllable representation spaces for dialogue.

1 Introduction

Learning numerical representations that preserve semantic similarity is a foundational problem in natural language processing and conversational artificial intelligence. From early distributional hypotheses to modern large language models (LLMs), the central premise has remained remarkably stable: words and concepts with similar meaning should be close in representation space. This principle underlies virtually all capabilities of conversational agents, including query understanding, response retrieval, dialogue state tracking, and personalized generation.

In conversational AI, the notion of similarity is broader than lexical semantics. Two user utterances "book a flight to NYC tomorrow" and "I need to reserve a plane ticket to New York for tomorrow" must be mapped to nearby vectors despite minimal lexical overlap. Similarly, system responses must be similar not only in topic but in style, persona, and pragmatic appropriateness. Representation learning therefore serves as the backbone for both understanding and generation.

This principle was first operationalized at scale by methods such as Word2Vec (Mikolov et al., 2013a,b), GloVe (Pennington et al., 2014), and later contextualized models like ELMo (Peters et al., 2018) and BERT (Devlin et al., 2019). These advances showed that geometry in embedding space captures rich linguistic and conceptual structure, enabling analogy solving ($e_{\text{king}} - e_{\text{man}} + e_{\text{woman}} \approx e_{\text{queen}}$), similarity search, and transfer to downstream tasks. For dialogue systems, this enabled a shift from rule-based intent matching to dense retrieval, where user intents could be matched via cosine similarity in $\mathbb{R}^d$.

However, moving from *observing* semantic structure to *controlling* it remains a major challenge. Recent work on representation engineering and concept detection demonstrates that while high-level concepts (e.g., sentiment, truthfulness, safety, toxicity, formality, persona, and dialogue acts) are linearly or non-linearly encoded in hidden states of LLMs, reliably detecting and understanding these concepts in representation space is non-trivial (Zou et al., 2023; Li et al., 2023). Representations are often polysemantic, entangled, and context-dependent (Elhage et al., 2022). A single neuron or direction may simultaneously encode multiple unrelated concepts, and a concept may be distributed across many dimensions.

This difficulty directly impacts our ability to steer conversational models toward a particular output type, which is a core requirement for deployable conversational AI. Steering — whether via activation editing, concept vectors, representation fine-tuning, or prompt-based intervention — requires not only that a concept exists in the space, but that it can be isolated, interpreted, and causally intervened upon without degrading other capabilities such as fluency, factuality, or helpfulness. In practice, we may want to steer a model toward more empathetic responses in a mental health support context, more concise answers in a task-oriented setting, or safer outputs when user queries are adversarial. Current

methods often achieve steering at the cost of quality degradation or poor generalization across prompts.

The gap between similarity preservation and steerability is particularly acute for conversational AI because dialogue is inherently multi-attribute and temporally dependent. A single response must simultaneously preserve semantic relevance to the user query, adhere to stylistic constraints (e.g., formal vs. casual), maintain a consistent persona, and satisfy safety policies. This requires a representation space where concepts are not only detectable via $\text{sim}(h_i, h_j)$ but also separable and manipulable via interventions $do(h')$. Moreover, steering at turn t influences turn $t+1$ via autoregression, creating compounding effects that are absent in static classification tasks.

This survey frames the evolution of representation learning as a progression toward this goal. We argue that while the field has largely solved similarity preservation, as evidenced by near-human performance on semantic textual similarity (STS) benchmarks, it has not yet solved concept disentanglement for controllable generation. We review historical methods that laid the geometric foundation, and state-of-the-art methods that attempt to address disentanglement and steering.

Our contributions are: (1) a unified formalism connecting count-based, static, and contextualized embeddings under the similarity preservation objective, (2) a taxonomy of modern methods based on their ability to support concept detection and steering, and (3) an outlook on open challenges for conversational AI, including causal evaluation and orthogonal concept encoding.

In Section 2, we provide background on distributional semantics and similarity-preserving embeddings, formalizing the objective $S_{human} \propto \text{sim}$. In Section 3.1, we survey historical methods for representation learning, from PPMI matrices and LSA to Word2Vec, GloVe, and FastText, highlighting their assumptions and limitations for steering. In Section 3.2, we survey state-of-the-art approaches, organized into contextualized encoders, sentence-level contrastive learning, LLM-derived embeddings, and representation engineering for steering. Finally, in Section 4, we discuss open challenges in concept detection and controlled steering and outline future directions.

2 Background

2.1 Distributional Hypothesis

The foundation of representation learning is the distributional hypothesis (Harris, 1954; Firth, 1957): words that occur in similar contexts tend to have similar meanings. Formally, the meaning of a word w can be inferred from the distribution $P(c|w)$ over its contexts c. The context c can be defined at various granularities: a fixed window of k words, a syntactic dependency, a sentence, or for conversational AI, an entire dialogue history $c = (u_1, s_1, ..., u_{t-1}, s_{t-1})$ where u_i and s_i are user and system turns.

2.2 From Counts to Dense Vectors

Early methods operationalized the distributional hypothesis via a co-occurrence matrix $M \in \mathbb{R}^{|V|\times|V|}$ where M_{ij} counts co-occurrences of word w_i and context c_j in a large corpus. To capture semantic association beyond raw frequency, Pointwise Mutual Information (PMI) is used:

$$\text{PMI}(w, c) = \log \frac{P(w, c)}{P(w)P(c)} \tag{1}$$

with positive PMI (PPMI) variant $\text{PPMI}(w, c) = \max(\text{PMI}(w, c), 0)$ to handle sparsity and negative associations (Levy and Goldberg, 2014). The row vector $M_{i:}$ then serves as a high-dimensional, sparse representation of w_i.

To mitigate this, dimensionality reduction techniques such as Singular Value Decomposition (SVD) are applied:

$$M \approx U_d \Sigma_d V_d^\top \tag{2}$$

where $U_d \in \mathbb{R}^{|V|\times d}$ contains the top-d left singular vectors. The embedding $E = U_d \Sigma_d^p$ with $p \in [0, 1]$ then yields dense vectors (Levy et al., 2015).

2.3 Similarity Preservation Objective

To obtain dense, low-dimensional vectors, we learn an embedding matrix $E \in \mathbb{R}^{|V|\times d}$ where $d \ll |V|$ and each word w is mapped to $e_w \in \mathbb{R}^d$. The standard measure is cosine similarity:

$$\text{sim}(w_i, w_j) = \cos(e_{w_i}, e_{w_j}) = \frac{e_{w_i}^\top e_{w_j}}{\|e_{w_i}\| \|e_{w_j}\|} \tag{3}$$

Under this view, the learning objective is to optimize E such that for a human-defined semantic similarity function S_{human}:

$$S_{human}(w_i, w_j) \propto \text{sim}(w_i, w_j) \tag{4}$$

A related property is relational similarity, where vector differences encode relations:

$$e_{\text{king}} - e_{\text{man}} + e_{\text{woman}} \approx e_{\text{queen}} \tag{5}$$

2.4 Learning Objectives for Conversational Relevance

In neural approaches, similarity preservation is achieved via self-supervised prediction tasks. For example, in Skip-gram (Mikolov et al., 2013a), we maximize the log-likelihood of context words given a center word:

$$\mathcal{J} = \frac{1}{T} \sum_{t=1}^{T} \sum_{-k \le j \le k, j \neq 0} \log P(w_{t+j}|w_t) \tag{6}$$

where $P(w_{t+j}|w_t) = \frac{\exp(e'^\top_{w_{t+j}} e_{w_t})}{\sum_{v\in V} \exp(e'^\top_v e_{w_t})}$ is parameterized by input and output embeddings. Levy and

Goldberg (Levy and Goldberg, 2014) showed this implicitly factorizes the shifted PPMI matrix.

For conversational AI, two additional desiderata emerge beyond Eq. 4: (1) *context sensitivity*: $e_w = f(w|c)$ must change with dialogue history, and (2) *disentanglement*: if a concept C is encoded, there should exist a direction v_C such that intervening $do(h + \alpha v_C)$ changes C without changing $C' \neq C$. Formally, we require:

$$\begin{aligned} p(C \mid h + \alpha v_C) > p(C \mid h) \quad \text{and} \\ p(C' \mid h + \alpha v_C) \approx p(C' \mid h) \quad \forall C' \neq C \end{aligned} \tag{7}$$

Static embeddings fail Eq. 7 due to superposition (Elhage et al., 2022). This motivates the methods in Sec. 3.2.

3 Methods

We organize representation learning methods by their ability to support concept detection and subsequent steering, with emphasis on conversational applications.

3.1 Historical Methods: Static Embeddings

Count-based models and early distributional methods. Building on Eq. 1, early work used SVD on PPMI matrices to obtain dense vectors $E = U_d \Sigma_d^p$ where $p \in [0, 1]$ is a smoothing parameter (Levy et al., 2015). Latent Semantic Analysis (LSA) and Hyperspace Analogue to Language (HAL) used similar ideas with different weighting.

Prediction-based models: Word2Vec, GloVe, FastText. Word2Vec introduced two efficient architectures (Mikolov et al., 2013a,b) that scaled distributional learning to billions of tokens. Skip-gram maximizes Eq. 6, while Continuous Bag-of-Words (CBOW) predicts a center word from context:

$$\mathcal{J}_{CBOW} = \frac{1}{T} \sum_{t=1}^{T} \log P(w_t \mid w_{t-k}, ..., w_{t-1}, w_{t+1}, ..., w_{t+k}) \tag{8}$$

The key innovation was negative sampling:

$$\mathcal{L}_{neg} = \log \sigma(e'^{\top}_{w_O} e_{w_I}) + \sum_{i=1}^{k} \mathbb{E}_{w_i \sim P_n(w)} [\log \sigma(-e'^{\top}_{w_i} e_{w_I})] \tag{9}$$

GloVe unifies both views by explicitly regressing on log-co-occurrences (Pennington et al., 2014):

$$\mathcal{J}_{GloVe} = \sum_{i,j} f(M_{ij})(e_i^{\top} e'_j + b_i + b'_j - \log M_{ij})^2 \tag{10}$$

where f is a weighting function $f(x) = \min((x/x_{max})^{\alpha}, 1)$. FastText (Bojanowski et al., 2017) extended this by representing $e_w = \sum_{g \in \mathcal{G}_w} z_g$, a sum of character n-gram embeddings.

Limitations for steering. Three limitations are critical: (1) *polysemy*, (2) *context insensitivity*, (3) *superposition*: with $d \ll |V|$, models must encode many more features than dimensions (Elhage et al., 2022). This directly violates Eq. 7.

3.2 State-of-the-Art Methods: Contextualized and Steering-Aware Representations

3.2.1 Contextualized encoders: ELMo, BERT, and beyond. ELMo (Peters et al., 2018) introduced deep contextualization via bidirectional LSTMs, where $e_w = \gamma \sum_{l=0}^{L} s_l h_l$. BERT (Devlin et al., 2019) replaced LSTMs with Transformers and optimized masked language modeling:

$$\mathcal{J}_{MLM} = -\sum_{m \in \mathcal{M}} \log P(w_m | c_{\setminus m}; \theta) \tag{11}$$

RoBERTa (Liu et al., 2019), DeBERTa (He et al., 2021) improved training dynamics. Probing studies showed that linear classifiers on h_l can detect concepts (Tenney et al., 2019). However, BERT embeddings are anisotropic (Ethayarajh, 2019).

3.2.2 Sentence-level contrastive learning. Methods like Sentence-BERT (Reimers and Gurevych, 2019) and SimCSE (Gao et al., 2021) use contrastive loss:

$$\mathcal{L}_{con} = -\log \frac{\exp(\text{sim}(h_i, h_i^+)/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(h_i, h_j^+)/\tau)} \tag{12}$$

This directly optimizes Eq. 4 and produces state-of-the-art results on MTEB (Muennighoff et al., 2023). TOD-BERT (Wu et al., 2020) and ConveRT (Henderson et al., 2020) fine-tune on response selection.

3.2.3 LLM-derived embeddings. Recent work shows decoder-only LLMs are effective embedding models via prompting or latent pooling (Jiang et al., 2024; BehnamGhader et al., 2024). LLM2Vec enables bidirectional attention in Llama and Mistral by removing causal masking.

3.2.4 Representation engineering and concept detection for steering. Let $h \in \mathbb{R}^d$ be an activation at a given layer l and token position t. A concept C is hypothesized to be encoded as a direction v_C:

$$h' = h + \alpha v_C \tag{13}$$

where α controls steering strength (Turner et al., 2023; Zou et al., 2023). Methods differ in how v_C is computed: Activation Addition (Turner et al., 2023), ITI (Li et al., 2023), ReFT, and CAVs (Kim et al., 2018).

The core difficulty is interference:

$$\text{sim}(v_C, v_{C'}) = \frac{v_C^{\top} v_{C'}}{\|v_C\| \|v_{C'}\|} \gg 0 \tag{14}$$

Methods like sparse autoencoders (SAEs) (Templeton et al., 2024; Cunningham et al., 2023) aim to disentangle superposed features:

$$h \approx \sum_{i=1}^{k} c_i f_i, \quad k \gg d, \quad \|c\|_0 \text{ small} \tag{15}$$

Conversational steering requires temporal consistency:

$$h'_{t+1} = f_\theta(h'_t, w_t), \quad w_{t+1} \sim P(\cdot|h'_{t+1}) \tag{16}$$

4 Conclusions

In this survey, we traced representation learning through the lens of a central challenge for conversational AI: moving from preserving semantic similarity to enabling controllable steering. Historical methods operationalized similarity as geometry and self-supervised prediction (Mikolov et al., 2013a,b; Pennington et al., 2014). However, static embeddings $e_w = f(w)$ fail on polysemy and entanglement (Elhage et al., 2022). State-of-the-art methods address this by reformulating representations as $e_w = f_\theta(w|c_1, ..., c_T)$ (Peters et al., 2018; Devlin et al., 2019) and by using contrastive learning (Reimers and Gurevych, 2019; Gao et al., 2021) to optimize MTEB (Muennighoff et al., 2023), with LLM-derived embeddings (Jiang et al., 2024; BehnamGhader et al., 2024). Beyond similarity preservation, current work seeks steerability as $h' = h + \alpha v_C$ (Turner et al., 2023; Zou et al., 2023) via SAEs (Templeton et al., 2024). Despite progress, non-orthogonality, temporal instability, and evaluation fragmentation remain. Future work should move toward causal benchmarks and orthogonal encoding.

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